

Total No. of Questions : 10]

SEAT No. :

P2071

[Total No. of Pages : 3

[5059]-677

B.E. (Information Technology)
ADVANCED GRAPHICS AND ANIMATION
(2012 Pattern) (Elective - III) (Semester - II)

Time : 2.30 Hours]

[Max. Marks : 70

Instructions to the candidates:-

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.*
- 2) Figures to the right indicate full marks.*
- 3) Neat diagrams must be drawn wherever necessary.*
- 4) Assume suitable data, if necessary.*

- Q1)** a) What are the different Spline specification methods? Explain with suitable diagram and examples. **[5]**
- b) Explain with mathematical model Bezier surface and B-Spline surface. **[5]**

OR

- Q2)** a) Explain the issues related to three dimensional display methods. Compare parallel projection and perspective projection method for 3-D objects. **[5]**
- b) Explain Regularized Boolean set operations with examples. **[5]**
- Q3)** a) Explain desirable properties in solid representation. **[5]**
- b) What is rendering? Explain Monte-Carlo method for rendering. **[5]**

OR

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- Q4)** a) Explain Phong's illumination model in detail. [5]
- b) What is tracing? Compare and Contrast Beam tracing and pencil tracing. [5]

- Q5)** a) What is blending? Explain in detail different blending operations performed in OpenGL. [8]
- b) Write a short note on "OpenGL primitives". [8]

OR

- Q6)** a) What is texture mapping? Explain any one texture mapping technique with implementation in OpenGL. [8]
- b) Explain any one Culling algorithm in detail. [8]

- Q7)** a) Explain briefly various real time animation techniques used in computer assisted animation. [8]
- b) Which are the different animation software's? Explain any one animation software in detail. [8]

OR

- Q8)** a) Why control hierarchy is required in animation? Explain various methods of controlling animation. [8]
- b) List and explain different animation languages. Explain features of animation languages. [8]

- Q9)** a) Explain virtual environment. How virtual reality help in good interactive design? [8]
- b) Enumerate different usability evaluation and testing techniques with respect to Virtual Reality. [10]

OR

- Q10)** a) What is Virtual Reality Modeling Language? List and explain different types of Virtual Reality? [8]
- b) What is the need of Virtual Reality? Explain in detail with real life example. [10]

